

Eduardo Abi Jaber

Professor of Applied Mathematics

École Polytechnique, Centre de Mathématiques Appliquées (CMAP)

eduardo.abi-jaber@polytechnique.edu | sites.google.com/view/abijabereduardo

Research

Stochastic systems with memory: Volterra processes, stochastic control and path signatures. Applications to volatility modelling, risk management, energy markets and sequential learning, developed through academic and industrial collaborations.

Academic appointments and education

Since 2025	Professor of Applied Mathematics , École Polytechnique, CMAP.
2022-2025	Assistant Professor of Applied Mathematics (tenure track) , École Polytechnique, CMAP.
2019-2022	Maître de conférences in Applied Mathematics , Université Paris 1 Panthéon-Sorbonne, Centre d'Économie de la Sorbonne.
2018-2019	Postdoctoral researcher , École Polytechnique, CMAP, with Mathieu Rosenbaum. DIM Math Innov fellowship.
2024	Habilitation à diriger des recherches (HDR) .
2016-2018	PhD in Applied Probability , Université Paris Dauphine, CEREMADE. Supervisors: Bruno Bouchard and Jean-David Fermanian; industrial co-supervisor: Camille Illand, AXA Investment Managers (CIFRE).
2015	Master's degree, M2 Modélisation aléatoire (M2MO) , Université Paris 7 Diderot.
2014	ENSAE diploma , quantitative finance and financial engineering.

Awards and distinctions

2019	AMIES PhD Prize , for doctoral research conducted in collaboration with industry (thesis completed in 2018).
2018	Bachelier Finance Society Junior Scholar Award , for work on multifactor approximation of rough volatility models.
2021	PEDR , Prime d'encadrement doctoral et de recherche.

Industrial partnerships

Research collaborations developed through CIFRE doctorates and postdoctoral contracts with **ENGIE Global Markets, AXA Investment Managers, BNP Paribas, CACIB and GEFIP**.

- **ENGIE Global Markets:** volatility and forward-curve modelling, joint historical and implied calibration, optimal trading and electricity storage. An energy-market model has been implemented internally.
- **AXA Investment Managers:** joint SPX-VIX modelling and calibration. The quintic volatility model has been implemented internally for risk-management applications.
- **BNP Paribas, CACIB and GEFIP:** CIFRE collaborations supporting doctoral research, including work on path signatures and financial modelling.

Doctoral and postdoctoral supervision

Current doctoral students

- **Ronan Pécheul** (since 2025), École Polytechnique. Co-supervisor: Clément Rey. **CIFRE: BNP Paribas.**
- **Dimitri Sotnikov** (since 2024), École Polytechnique. Co-supervisor: Charles Bertucci. **CIFRE: ENGIE Global Markets.** Recipient of the European Best Quant Finance Master's Thesis Award 2024.
- **Elie Attal** (since 2024), École Polytechnique. Co-supervisor: Mathieu Rosenbaum. Recipient of the Bloomberg Quantitative Finance PhD Fellowship.
- **Maxime Guellil** (since 2024), Université Paris Cité. Co-supervisor: Stéphane Crépey. **CIFRE: CACIB.**

Completed doctorates

- **Louis-Amand Gérard** (2022-2025), Université Paris 1 Panthéon-Sorbonne. Co-supervisor: Olivier Guéant. **CIFRE: GEFIP.** Research on path signatures.
- **Nathan De Carvalho** (2021-2025), Université de Paris. Co-supervisor: Huyên Pham. **CIFRE: ENGIE Global Markets.** Thesis: *Lifting Energy Markets: from Volatility Modeling to Optimal Trading*. Recipient of the Bachelier Finance Society Junior Scholar Award 2024 and the Bruti-Liberati Prize 2025. **Current position:** Quant at CFM.
- **Shaun Li** (2021-2024), Université Paris 1 Panthéon-Sorbonne. Co-supervisor: Bernard de Meyer. **CIFRE: AXA Investment Managers.** Thesis: *The Quintic Volatility Model for SPX and VIX: A 360 Tour*. **Current position:** Quant at Morgan Stanley.

Postdoctoral researchers and other mentoring

- **Francesca Primavera** (since 2025).
- **Alessandro Bondi** (2023-2024), École Polytechnique; co-supervision with Sergio Pulido. **Current position:** Tenure-track researcher at Luiss University, Rome.
- Visiting doctoral researcher: **Édouard Motte** (2025). Research collaboration with **Enzo Miller** (2019-2021; PhD supervisor: Huyên Pham), resulting in three joint articles.
- Supervision of more than 30 master's research projects and internships, including projects with ENGIE, BNP Paribas, HSBC and GEFIP.

Editorial appointments and scientific responsibilities

Associate editor, since 2026: *Mathematical Finance; Finance and Stochastics; International Journal of Theoretical and Applied Finance.*

Academic leadership: Head of the third-year Applied Mathematics track at École Polytechnique (150 students; since 2024); responsibility for internship modules and practitioner seminars (since 2022). Director of studies, M2 IRFA, Université Paris 1 (2020-2022).

Scientific organisation: Séminaire Bachelier, École Polytechnique representative (2022-2026); London-Paris Bachelier Workshops (2024, 2025); CMAP 50th anniversary conference (2024). Organiser of memory-themed mini-symposia at SIAM and Stochastics & Computational Finance (2025), and the Bachelier World Congress and ICCF (2026).

Evaluation and service: 11 doctoral examination committees, as referee or examiner; faculty selection committees at Université Paris Dauphine (2022) and Université Paris Cité (2023). More than 85 journal referee reports.

Selected publications and preprints

Research output: 47 research papers and preprints.

Reference numbers match the accompanying bibliography; this selection covers foundations, numerical methods, control and applications.

Foundations and numerical methods

[3] Affine Volterra Processes

E. Abi Jaber, M. Larsson and S. Pulido. *The Annals of Applied Probability*, 29(5), 3155-3200, 2019.

[4] Multifactor Approximation of Rough Volatility Models

E. Abi Jaber and O. El Euch. *SIAM Journal on Financial Mathematics*, 10(2), 309-349, 2019. Bachelier Finance Society Junior Scholar Award 2018.

[7] Weak Existence and Uniqueness for Affine Stochastic Volterra Equations with L^1 -Kernels

E. Abi Jaber. *Bernoulli*, 27(3), 1583-1615, 2021.

[23] Polynomial Volterra Processes

E. Abi Jaber, C. Cuchiero, L. Pelizzari, S. Pulido and S. Svaluto-Ferro. *Electronic Journal of Probability*, 29, 1-37, 2024.

Stochastic control and decisions with memory

[8] Linear-Quadratic Control for a Class of Stochastic Volterra Equations: Solvability and Approximation

E. Abi Jaber, E. Miller and H. Pham. *The Annals of Applied Probability*, 31(5), 2244-2274, 2021.

[16] Optimal Liquidation with Signals: The General Propagator Case

E. Abi Jaber and E. Neuman. *Mathematical Finance*, 35(4), 841-866, 2025.

Financial and energy-market applications

[17] Joint SPX-VIX Calibration with Gaussian Polynomial Volatility Models: Deep Pricing with Quantization Hints

E. Abi Jaber, C. Illand and S. Li. *Mathematical Finance*, 35(2), 470-519, 2025.

[29] Heath-Jarrow-Morton Meets Lifted Heston in Energy Markets for Joint Historical and Implied Calibration

E. Abi Jaber, S. Bruneau, N. De Carvalho, D. Sotnikov and L. Tur. *Quantitative Finance*, accepted for publication.

Path signatures and learning

[21] Signature Volatility Models: Pricing and Hedging with Fourier

E. Abi Jaber and L.-A. Gérard. *SIAM Journal on Financial Mathematics*, 16(2), 606-642, 2025.

[38] Exponentially Fading Memory Signature

E. Abi Jaber and D. Sotnikov. Preprint, 2025.

Selected invited lectures and minicourses

More than 120 conference and seminar presentations to academic and professional audiences. Selected invitations are listed below.

Plenary lectures, keynote and special invited lecture

- 2026** **Bloomberg Quant BBQ**, New York, February. Keynote: *Path-Signatures: Memory and Stationarity*.
- 2026** **Actuarial and Financial Mathematics Conference**, Brussels, February. Plenary: *Volatility Dynamics and Memory: from the Quintic Model to Signatures*.
- 2026** **Advances in Financial Mathematics**, Paris, January. Plenary: *Efficient Simulation of Affine Volterra Processes: from Heston to Hawkes*.
- 2026** **Winter School on Mathematical Finance**, Soesterberg, January. Special invited lecture on path signatures and memory.
- 2025** **Vienna Congress on Mathematical Finance**, July. Plenary on volatility dynamics and memory.

Minicourses and practitioner presentations

- 2024** **Research in Options**, Rio de Janeiro. Minicourse on Volterra processes in quantitative finance.
- 2023** **Stochastic Methods in Finance and Physics**, Crete. Minicourse on Volterra processes in finance.
- 2023** **KAUST**, Saudi Arabia. Nine-hour minicourse on non-Markovian models in finance, with Christian Bayer.
- 2025** **CBOE RMC Quant Conference**, Chicago. Invited talk on volatility dynamics and the quintic model.
- 2023-2025** **Practitioner seminars**: Goldman Sachs (2023); BNP Paribas, Crédit Agricole and Morgan Stanley (2025).

Teaching and course development

École Polytechnique and Sorbonne Université, M2 Probabilités et Finance

- *Memory in Quantitative Finance: Volterra Processes and Path Signatures* (15 hours). Course created in 2023; taught annually, 2023-2026.
- *Stochastic Modelling and Derivatives* (21 hours), 2025.

École Polytechnique and HEC, MSc Data Science for Finance: Deep Learning in Finance (24 hours), 2024-2025.

Université Paris 1 and ENSAE, 2019-2022: lectures in machine learning, calibration, numerical methods, mathematical finance and market risk; tutorials in probability, analysis and insurance mathematics.

Executive and professional education: stochastic calculus and control, École Polytechnique (2023-2025); volatility and credit risk, BNP Paribas model-inspector training (2023); financial engineering courses, UM6P and École Polytechnique (2023).

Selected research funding

- 2022** **PHC AMADEUS**, Campus France. *Polynomial Volterra Processes and Their Applications in Finance*, with Christa Cuchiero, Sergio Pulido and Sara Svaluto-Ferro.
- 2018-2019** **DIM Math Innov postdoctoral fellowship**, École Polytechnique.